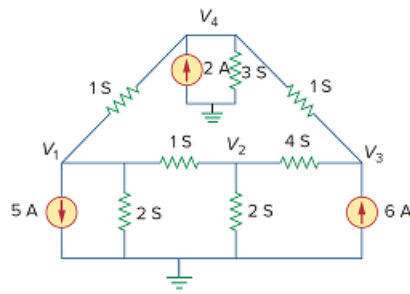


Problem

Use *MATLAB* to solve for the node voltages in the circuit of Fig. 3.78.

Figure 3.78:



Step-by-step solution

Step 1 of 2

Refer to the circuit diagram in Figure 3.78 in the textbook.

Apply Kirchhoff's current law at node 1.

$$2v_1 + 1(v_1 - v_2) + 1(v_1 - v_4) + 5 = 0$$

$$4v_1 - v_2 - v_4 = -5 \quad \dots\dots (1)$$

Apply Kirchhoff's current law at node 2.

$$1(v_2 - v_1) + 2v_2 + 4(v_2 - v_3) = 0$$

$$-v_1 + 7v_2 - 4v_3 = 0 \quad \dots\dots (2)$$

Apply Kirchhoff's current law at node 3.

$$4(v_3 - v_2) + 1(v_3 - v_4) = 6$$

$$-4v_2 + 5v_3 - v_4 = 6 \quad \dots\dots (3)$$

Apply Kirchhoff's current law at node 4.

$$1(v_4 - v_1) + 3v_4 + 1(v_4 - v_3) = 2$$

$$-v_1 - v_3 + 5v_4 = 2 \quad \dots\dots (4)$$

Step 2 of 2

Write four equations in matrix form.

$$\begin{bmatrix} 4 & -1 & 0 & -1 \\ -1 & 7 & -4 & 0 \\ 0 & -4 & 5 & -1 \\ -1 & 0 & -1 & 5 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix} = \begin{bmatrix} -5 \\ 0 \\ 6 \\ 2 \end{bmatrix}$$

Solve equations (1),(2),(3) and (4).

```
>> A=[4 -1 0 -1; -1 7 -4 0; 0 -4 5 -1;-1 0 -1 5];
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>> B=[-5;0;6;2];
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>> X= inv(A)*B;
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X =

```
   -0.7708    1.2093    2.3090    0.7076
```

Voltage v_1 is -0.7708 V

Voltage v_2 is 1.2093 V

Voltage v_3 is 2.3090 V

Voltage v_4 is 0.7076 V